

Curriculum Vitae

PERSONAL INFORMATION

Walter Lucas, Ph.D.
University of Brasília
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Birth date: 15 May 1992
Place of birth: Brasília, Brazil
Nationality: Brazilian

EDUCATION

- | | |
|-------------|---|
| 2019 - 2026 | Ph.D., Computer Science
University of Brasília, UnB, Brazil
Thesis: A Socio-Technical Grounded Theory about Source Code Rejuvenation
Advisor: Rodrigo Bonifácio de Almeida
Co-Advisor: João Alexandre Baptista Vieira Saraiva |
| 2017 - 2019 | Master.Sc., Computer Science
University of Brasília, UnB, Brazil
Thesis: Analysis of the impact on the understanding of Java programs with the adoption of lambda expressions
Advisor: Rodrigo Bonifácio de Almeida |
| 2011 - 2015 | Bachelor's Degree, Information System
University Center of Patos de Minas, UNIPAM, Brazil
Thesis: UNIEXPO: platform for dissemination of extension projects
Advisor: Jean Lucas de Souza |

PROFESSIONAL AND ACADEMIC EXPERIENCES

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| 2024 - Current | Fóton Informática S.A. Position: Java Developer Analyst
Project: Conduction and refinement of requirements, legacy system modernization, troubleshooting, refactoring, and static analysis using SonarLint and Fortify, alongside automated testing with JUnit and Mockito.
Research topic: Software Engineering, Static Analysis, Code Quality |
| 2023 - 2024 | Analysis of the Impact of ESV on the Social Reintegration Process (Phase 2) Position: Developer and Product Owner / Researcher
Project: Developed new features to facilitate information sharing among partner institutions, coordinating backend, web frontend, and mobile teams. Integrated updates including Expo migration, React Native, chatbots, and robust authorization systems.
Research topic: Software engineering, Privacy, Mobile Applications
Supervisor: Rodrigo Bonifácio de Almeida |

2023 - 2023	INESC TEC International Visiting Researchers Programme – 2022 Edition Position: Researcher Project: Studied development practices of KDE contributors regarding modern C++ feature adoption across GitHub repository histories. Research topic: Software engineering, C++ evolution Supervisor: João Saraiva - University of Minho - Portugal
2020 - 2022	Foundation for Scientific and Technological Enterprise - Finatec (Phase 1) Position: Data Engineer / Developer and Researcher Project: Exploring the Use of privacy and user-centric requirement elicitation techniques in a social inclusion context, developing ETL solutions for the Virtual Social Office (ESV). Research topic: Software engineering, Data Management Supervisor: Rodrigo Bonifácio de Almeida
2018 - 2020	Research and Development, Department of Electrical Engineering - UNB Position: Data Engineer / Software Developer Project: Applied Research in Strategic Planning, Corporate Risk Management and Corporate Information Management within the Ministry of Planning, Budget and Management (DIPLA/MP) Research topic: Software engineering, Business Intelligence and Analytics Supervisor: Rafael Tímoteo de Sousa Junior
2017 - 2018	Foundation for Scientific and Technological Enterprise - Finatec Position: Software Developer Project: Research to Validate Software Development Practices and Methods of Software Development for the Brazilian Army (PromiseEB) Research topic: Software engineering, Software Visualization Supervisor: Rodrigo Bonifácio de Almeida
2016 - 2017	Mídia Proativa Position: PHP Developer Project: Development and management of e-commerce and web systems, including the Fenamilho 2016 management platform. Research topic: Web development, PHP frameworks
2015 - 2015	MAXPHONE Position: Systems Analyst Project: Development of web solutions using PHP/CakePHP, database optimization, and data-driven sales strategies using Google Analytics. Research topic: E-commerce, Analytics

PUBLICATIONS

Journal Publications

1. MENDONÇA, W. L. M., FORTES, J., LOPES, F. V., MARCÍLIO, D., DE ALMEIDA, R. B., CANEDO, E. D., LIMA, F., AND SARAIVA, J. Understanding the impact of introducing lambda expressions in java programs. *Journal of Software Engineering Research and Development* 8 (2020), 7–1
2. LUCAS, W., CARVALHO, F., NUNES, R. C., BONIFÁCIO, R., SARAIVA, J., AND ACCIOLY, P. Embracing modern c++ features: An empirical assessment on the kde community. *Journal of Software: Evolution and Process* 36, 5 (2024), e2605

Conferences

1. LUCAS, W., BONIFÁCIO, R., CANEDO, E. D., MARCÍLIO, D., AND LIMA, F. Does the introduction of lambda expressions improve the comprehension of java programs? In *Proceedings of the XXXIII Brazilian Symposium on Software Engineering* (2019), pp. 187–196
2. DE MENDONÇA, W. L. M., COSTA, P. H. T., CANÇADO, E. C. R., LIMA, F., CANEDO, E. D., BONIFÁCIO, R., AND AMARAL, L. H. V. From dusk till dawn: Reflections on the impact of covid-19 on the development practices of a r&d project. In *Proceedings of the 34th Brazilian Symposium on Software Engineering* (2020), pp. 596–605
3. DA FROTA, F. V. L., DE MENDONÇA, W. L. M., DA SILVA, D. A., DE SOUZA FERNANDES, M. A., BARBOSA, N. F., AND CANEDO, E. D. Utilização de algoritmos genéticos em gerações de dados de teste de software com agrupamento de genes e redução de espaço de busca. *WWW/INTERNET 2019*, 87

Extended Abstract / Short Papers

1. DANTAS, R., CARVALHO, A., MARCÍLIO, D., FANTIN, L., SILVA, U., LUCAS, W., AND BONIFÁCIO, R. Reconciling the past and the present: An empirical study on the application of source code transformations to automatically rejuvenate java programs. In *2018 IEEE 25th international conference on software analysis, evolution and reengineering (SANER)* (2018), IEEE, pp. 497–501
2. LUCAS, W., BONIFÁCIO, R., AND SARAIVA, J. Understanding the motivations, challenges, and practices of software rejuvenation. In *2023 IEEE International Conference on Software Maintenance and Evolution (ICSME)* (2023), pp. 611–616

HONORS, AWARDS & SCHOLARSHIPS

2019	Best paper award in Proceedings of the XXXIII Brazilian Symposium on Software Engineering (SBES).
2016	Unideias, Patos de Minas University Center.
2015	7th Entrepreneurship Award, Patos de Minas University Center.

LICENSES & CERTIFICATIONS

2026	AWS Certified AI Practitioner - Amazon Web Services (AWS)
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LANGUAGES

Portuguese (Native)
English (Advanced, TOEFL ITP: 443)
Spanish (Intermediate)